Vibe Coding for Open Source

- by Nick Hao
- for Open Source Hong Kong

- 2025/11/13

Senior Analyst
HKPC - Hong Kong Productivity Council · Full-time
Nov 2022 - Present · 3 yrs 1 mo

My main role is to provide Digital Enablement, AI, and IoT services with Security by Design. My clients include SMEs, large corporations, and government bodies in Hong Kong.

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Consultant, Industry 4.0+
New Industrialisation Division
Ref: NID/10657 → Position Reference Code

Job Description

Who we are looking for:
- Creativity in new ideas and future focus vision.
- Collaboration with colleagues and the community at large.
- Work Smart, Play Smart!

Agenda (3 parts, 5 live demos)

1. Vibe Coding allows Non-IT people contributing to Open Source

- Demo 1: Use Human language + Browser to create a website, and Sync the Code to Github
- Demo 2: Use AI Browser to generate Data Analysis Report with HTML, and Deploy it on GitHub for free

2. Vibe Coding allows IT people to create projects with languages they haven't learned before

- Demo 3: Use AI to implement Dark Mode for an HTML Page
- Demo 4: Use AI to generate Python code for working with Excel

3. My open-source project by Vibe Coding

- Demo 5: My open-source project for searching Hong Kong events

Part 1. Vibe Coding for Non-IT people

The times have changed

English in China (Human Language)

China joined WTO (World Trade Organization)
And rapid globalization began

Majored in English to master English

A coder can also speak English

Past → Future

Code (Computer Language)

LLM was invented
And AI can now code better than humans

Majored in IT to code

A lawyer can also code with AI

Middle school students in HK are learning code and AI

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Demo 1: Use Human language + Browser to create a website, and Sync the Code to Github

Only need a browser, even a **child** can create a website using Human Language.

Lovable	https://lovable.dev/	Lovable
Bolt	https://bolt.new	bolt
Replit	https://replit.com	replit
V0	https://v0.app/	v0

English Prompt

Create a brightly colored, cartoon-style personal website for elementary student Nick. Please include the following features and content:

A public page
- A cute homepage displaying the website name and basic welcome message for children (“Little Island Elementary School, Grade 3, Class 04, boy”) along with a profile picture
- An “My Interests” section listing 4 interests: swimming, skateboarding, reading, and gaming, each represented by an adorable icon
- A message board section where anyone can leave a message for Nick, and optionally note their own name
- The page must automatically adapt for mobile display
- In the upper left corner of the homepage, show a large QR code of the current browser page URL

The public page's upper right corner features an “Admin” button. After entering username and password, users access a backend admin page
- Set initial username as "admin@nick.com" and initial password as "123456", with an option to change the password
- The admin can update the profile picture, supporting local image upload
- The admin can delete any message they do not like

Lovable generated project

nicky-bright-bytes

Preview

Share Upgrade Publish

為小學生Nick創建一個色彩鮮明、內容簡單的個人網站。請包含以下內容與功能:

一個公開頁面
- 一個可愛的首頁，顯示網站名稱和歡迎小朋友的基本訊息(小島中學3年級04班,男仔)與頭像，用明亮卡通風格配色
- 一個'我的興趣'區塊，列出4個興趣，游泳，滑板與打機，每個用可愛圖標表示
- 一個留言專區，任何人都可以給Nick小朋友留言，並可選備註自己的名字
- 頁面需自動適應手機顯示

一個後台頁面
- 初始密碼為 "123456"
- 可以看到所有留言
- 可以更換自己的頭像

Lovable

Thought for 22 seconds

我將為 Nick 小朋友創建一個充滿童趣、色彩鮮明的個人網站!

設計靈感:

- 參考兒童友好的網站設計，如 National Geographic Kids
- 使用明亮的藍色、橙色、綠色、紫色配色
- 卡通風格圖標和圓潤設計
- 柔和的陰影和動畫效果

功能實現:

0 free credits remaining today

Upgrade Now

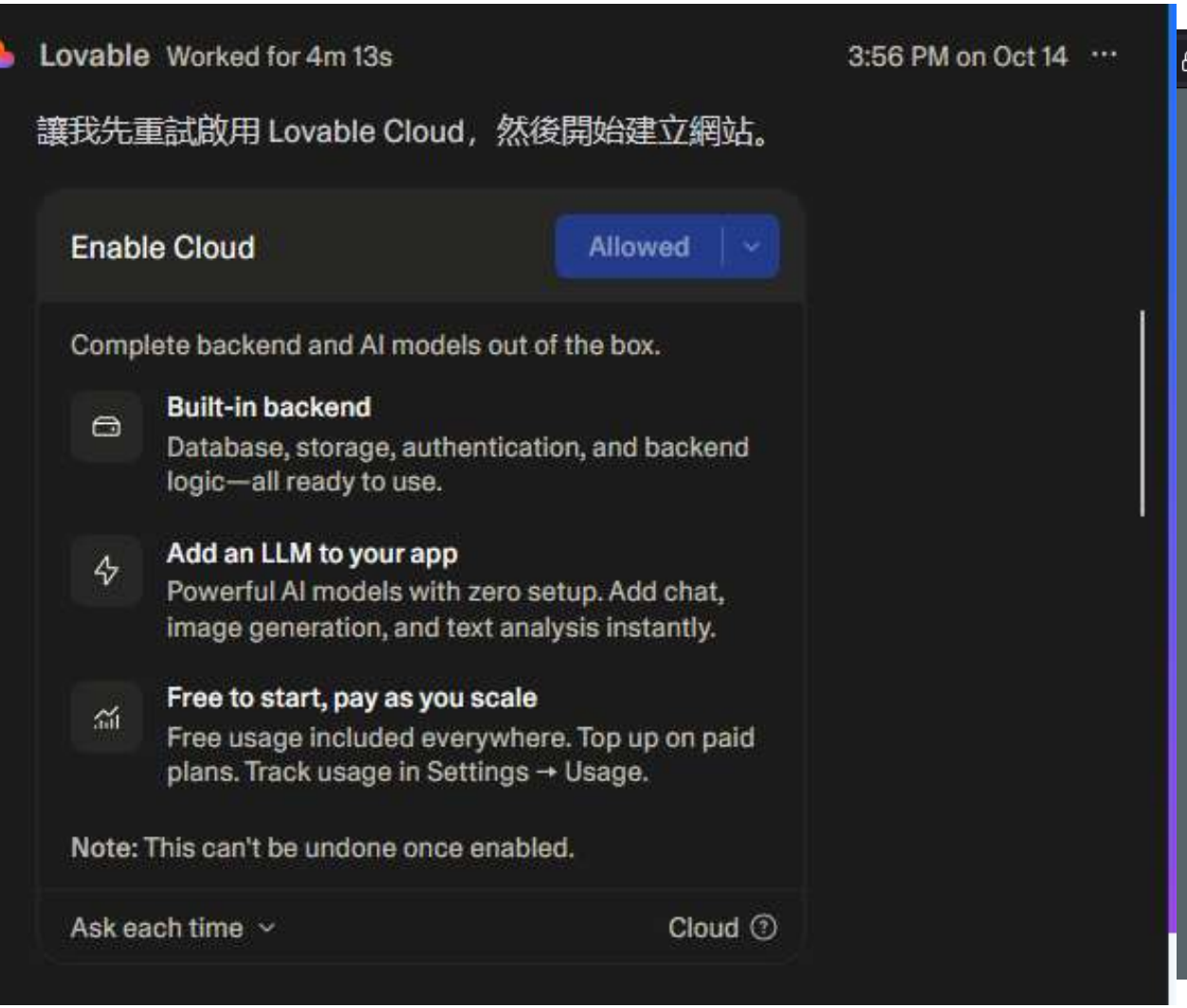
Ask Lovable...

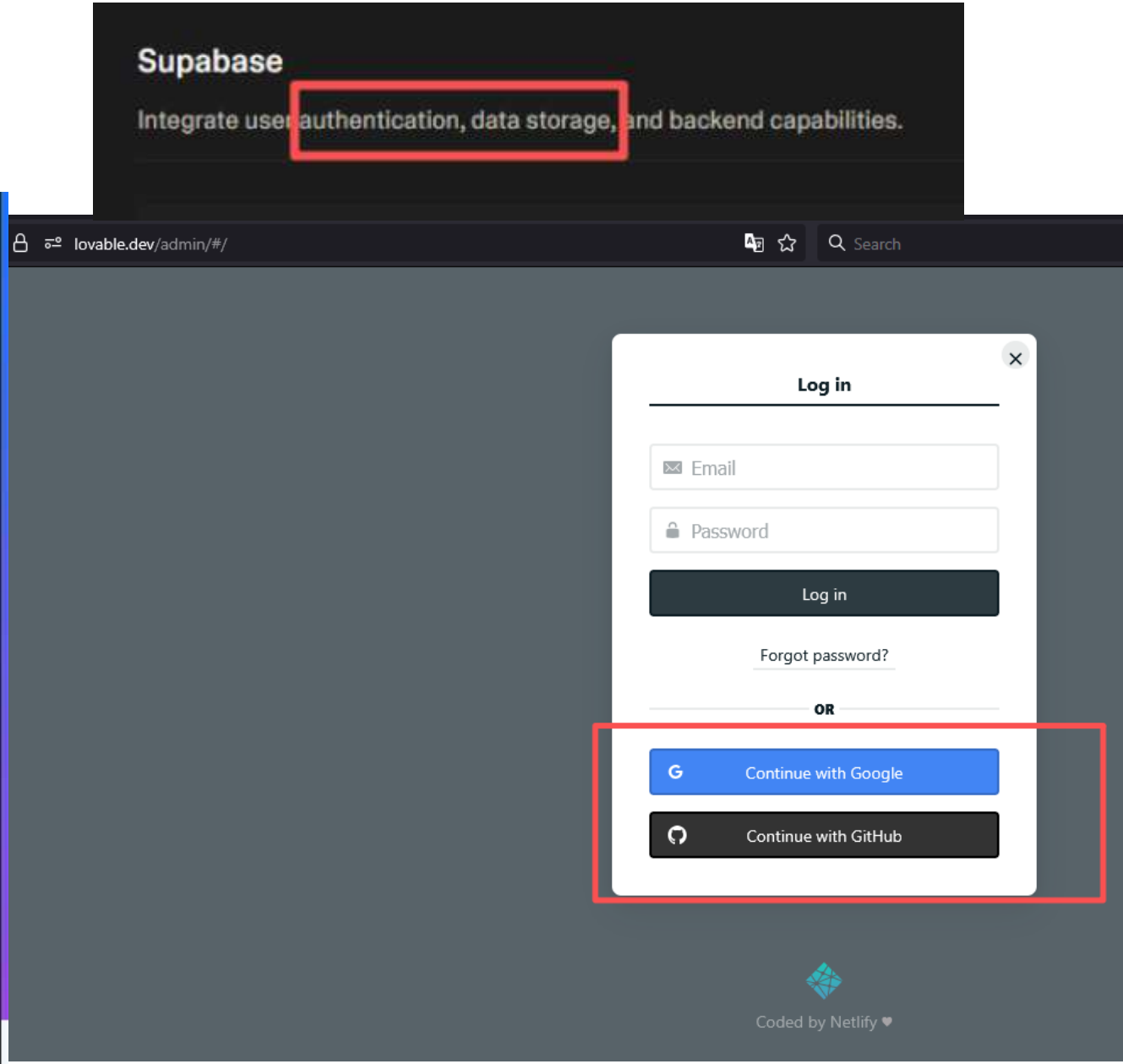
Edit

Chat

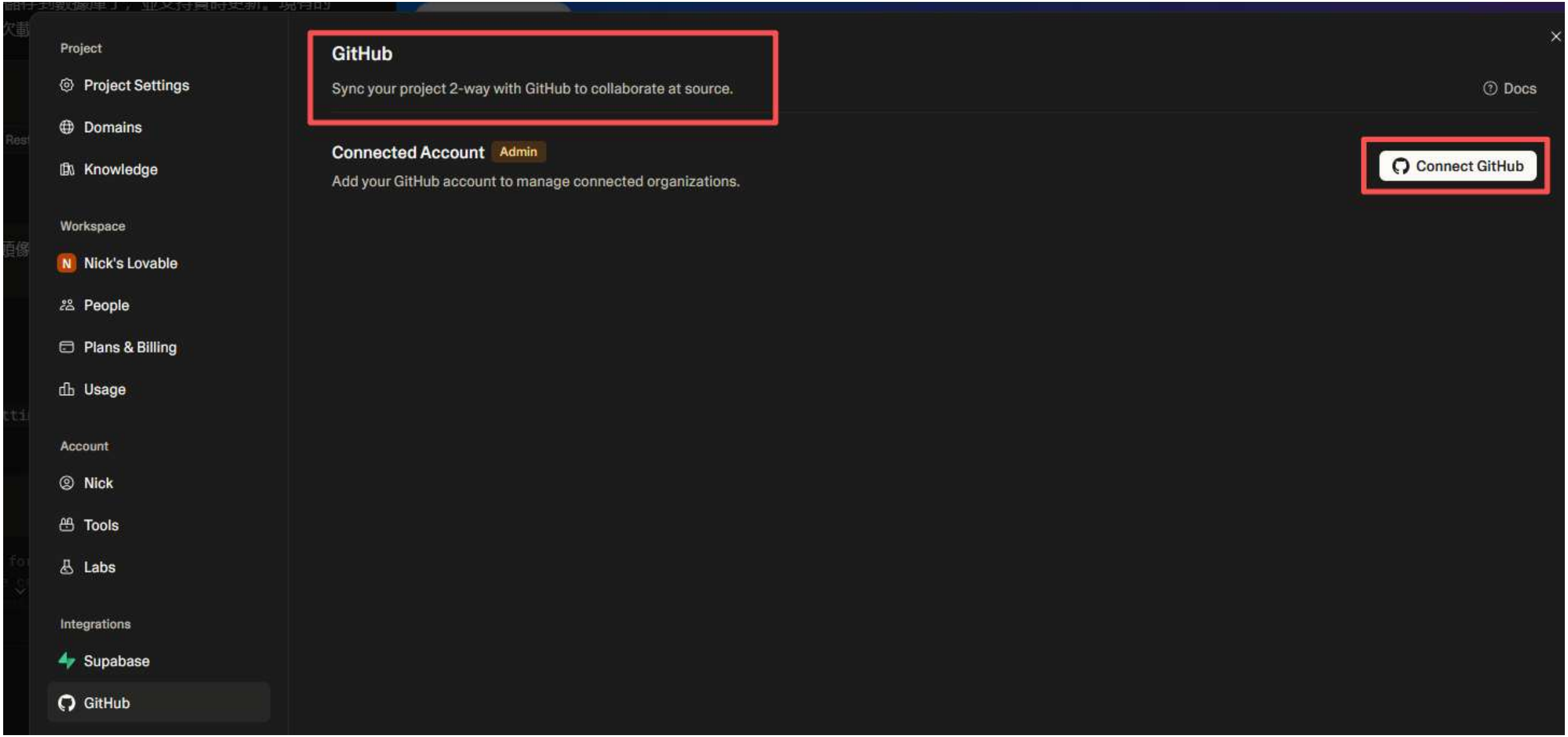
Lovable integrated：

- Database
- LLM
- Authentication
- others





Sync the code to you Github to enable it to be an Open Source project

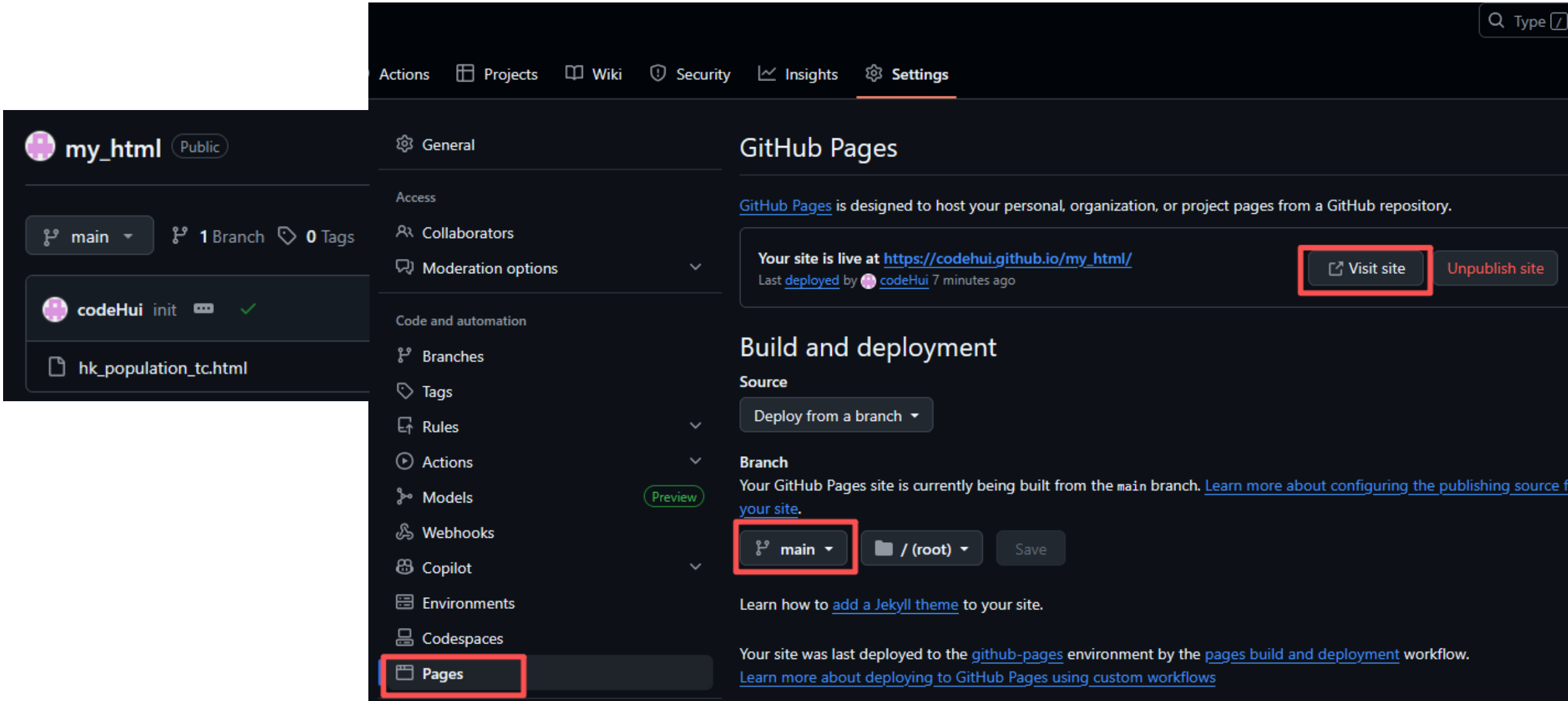


Demo 2: Use AI Browser to generate Data Analysis Report with HTML, and Deploy it on GitHub

Use AI Browsers like "Comet" or "Doubao" to generate a Data Analysis Report with HTML, and download the HTML file

Open-source and deploy the HTML

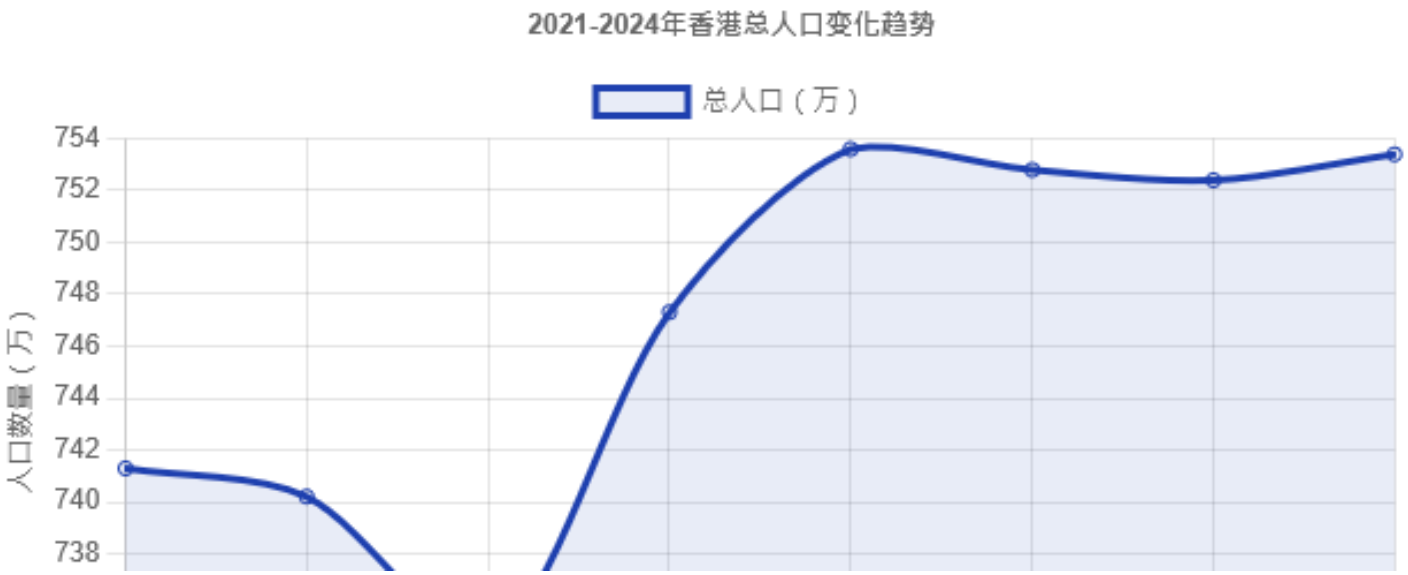
1. Upload the HTML to GitHub as an open-source project
2. Deploy it via GitHub Pages for free (Choose a branch)



香港人口发展分析

概述 总人口 性别结构 年龄分布 老龄化趋势 结论建议

2021-2024年香港总人口呈现波动增长趋势，从2021年中的741.3万增长至2024年中的752.4万，年均增长率约0.4%。



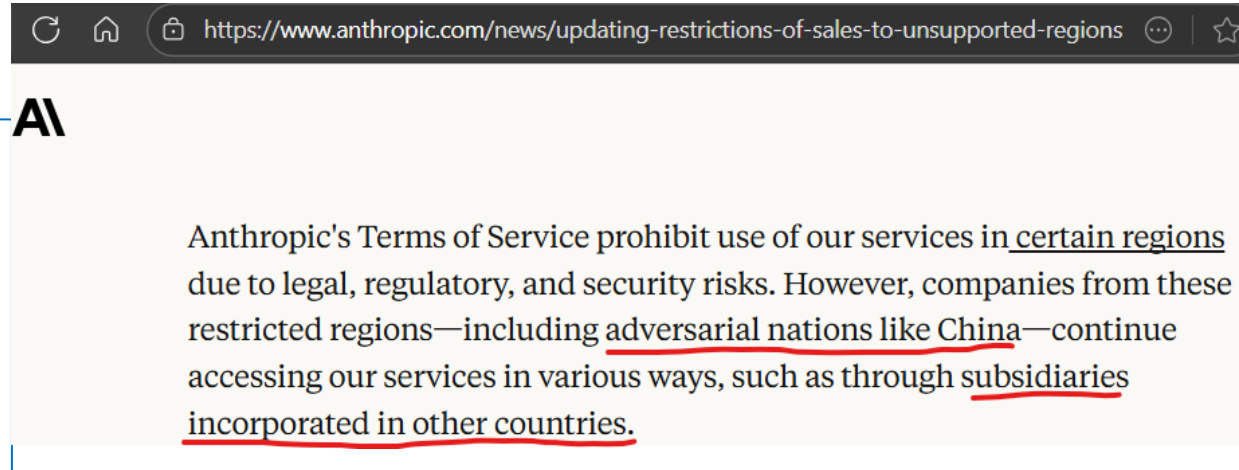
Part 2. Vibe Coding for IT people

Vibe Coding tools for IT people

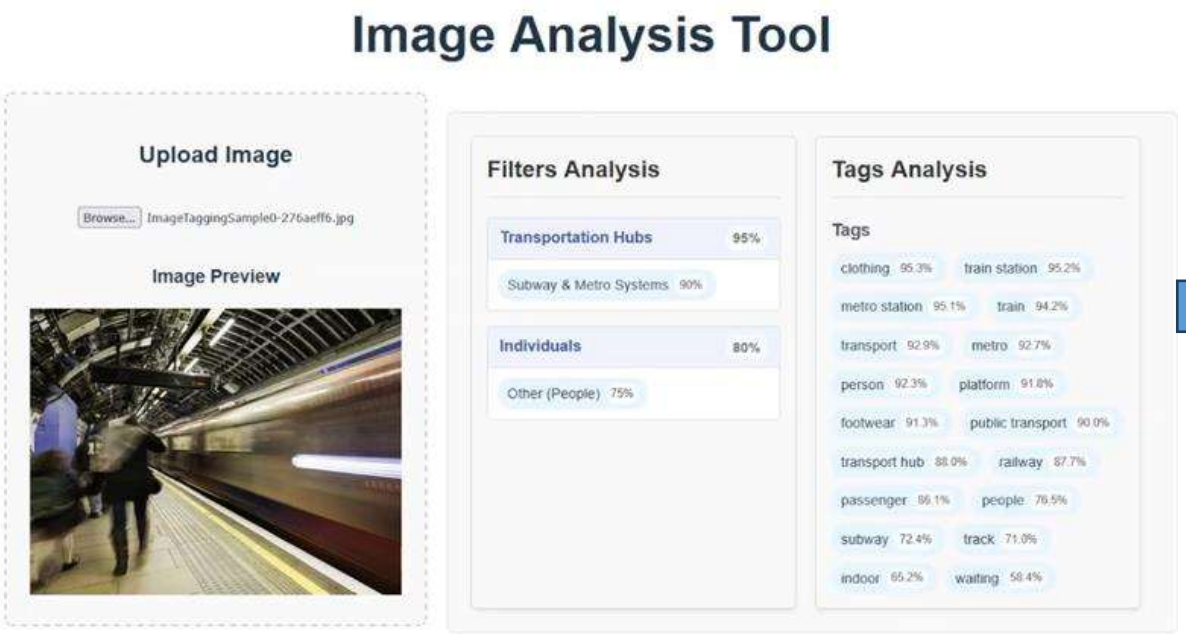
Product	Company	Product Type	Website
GitHub Copilot	GitHub (Microsoft)	IDE Extension (well supported by VS Code)	https://github.com/features/copilot
Cursor	Anysphere	AI IDE	https://cursor.com/
Amazon Q	Amazon	IDE Extension	https://aws.amazon.com/q/
Kiro	Amazon	AI IDE	https://kiro.dev/
Windsurf	Windsurf	IDE Extension	https://windsurf.com/editor
Firebase Studio	Google	Cloud AI IDE	https://firebase.studio/
Trae	ByteDance	AI IDE, IDE Extension	China Version: https://www.trae.com.cn/ International Version: https://www.trae.ai/
通义灵码	Alibaba	IDE Extension	https://lingma.aliyun.com/
CodeBuddy	Tencent	AI IDE	https://copilot.tencent.com/

Recommendation for GitHub Copilot over Cursor:

- Extensions support outside of VS Code (e.g., in IntelliJ IDEA, on the GitHub website)
- Backed by a major company, Microsoft
- Your code is on GitHub, more secure



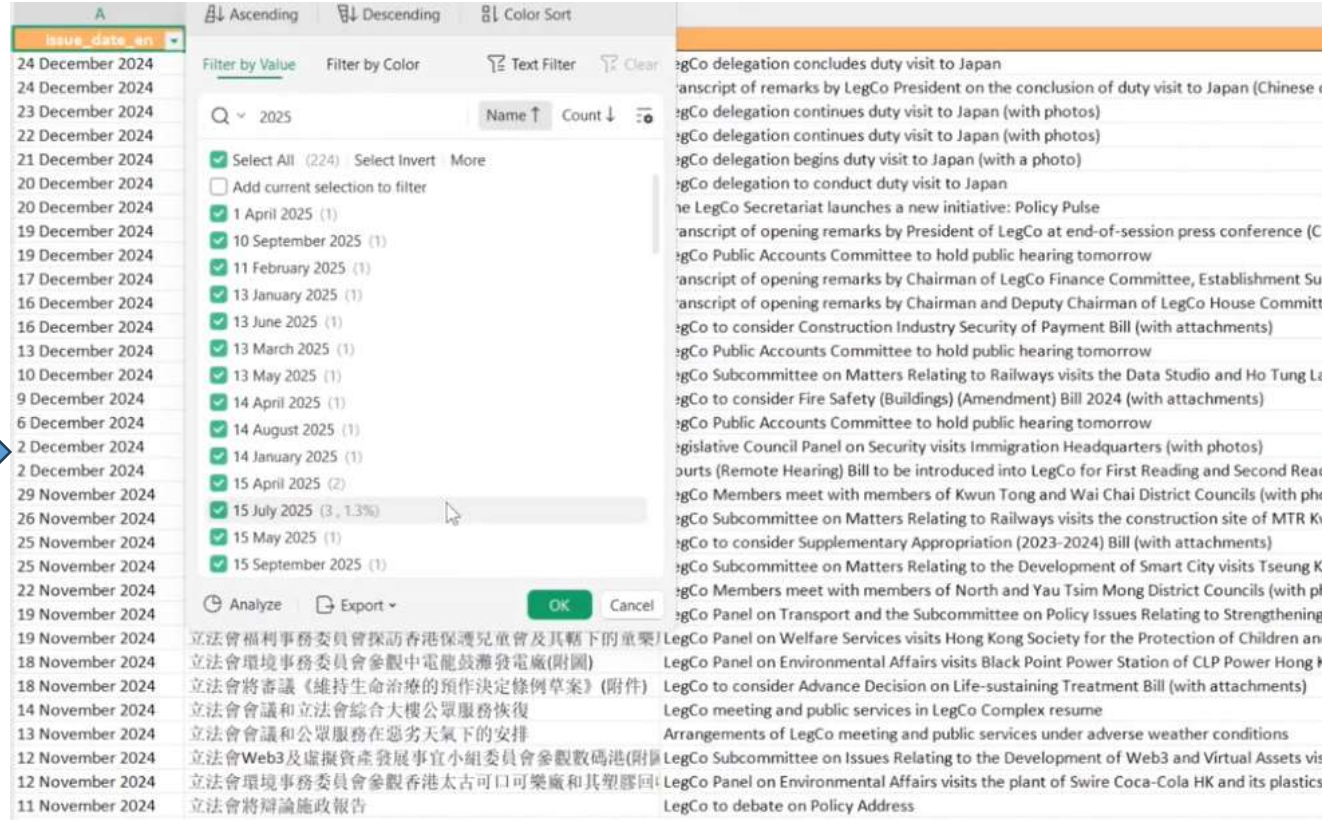
Demo 3: Use AI to implement Dark Mode for an HTML Page



Demo 4: Use AI to generate Python code for working with Excel

Demo Task:

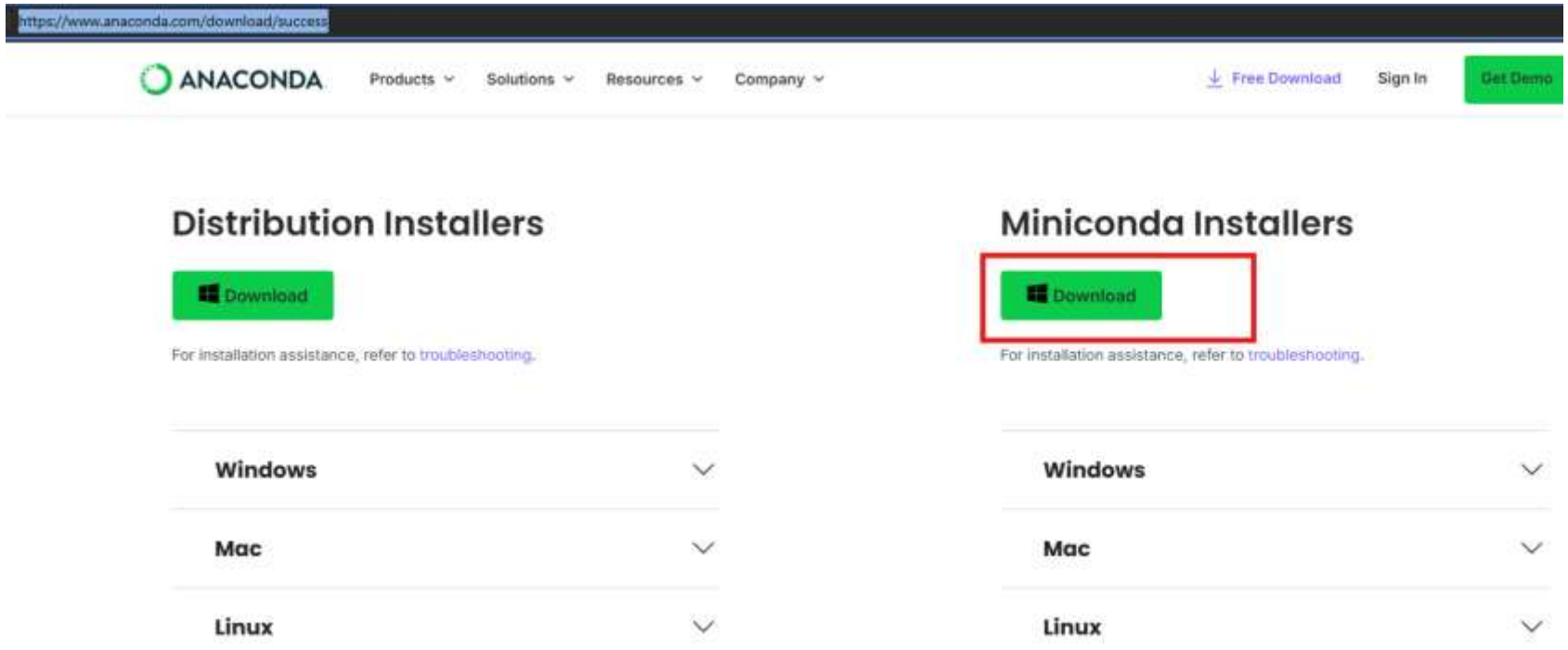
On the left is the Legislative Council's press release website (https://app4.legco.gov.hk/press/en/press.aspx). The goal is to copy the publication date, Chinese and English titles of all press releases from the past five years into an Excel file.



Start with Python as a layman - Install Miniconda

Miniconda is a Python environment management tool.
By installing Miniconda, you don't need to install a standalone Python environment directly.
Miniconda allows you to manage multiple Python environments—for example, one project can use Python 3.9 while another uses Python 3.11.

Miniconda download link: https://www.anaconda.com/download/success



English Prompt for Python Beginner

Please generate a convert_press_to_excel.py script, and use a conda environment.yml file for dependency management.

Below is the requested URL:
https://app4.legco.gov.hk/press/get_press.aspx?pageno=1&record_per_page=1000&current_term_f...
It returns data in the following JSON format.
'''

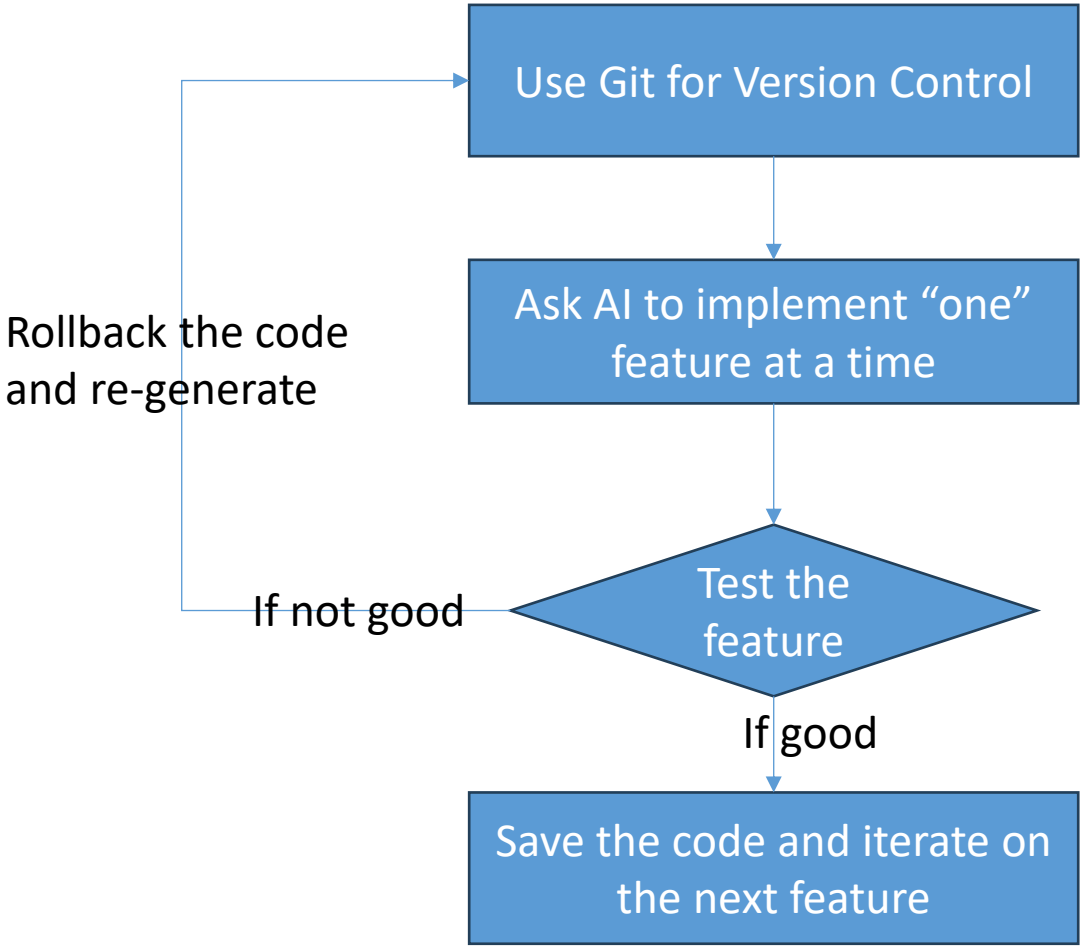
```
[{
        "date_group": "2025年10月",
        "date_group_en": "October 2025",
        "pdf_link": "//www.legco.gov.hk/tc/open-legco/press/yr2025/pr20251006-1.html",
        "pdf_link_en": "//www.legco.gov.hk/en/open-legco/press/yr2025/pr20251006-1.html",
        "intRow": "1",
        "intTotalHits": "81",
        "id": "19820",
        "type_id": "1",
        "issue_date": "2025年10月6日",
        "issue_date_en": "6 October 2025",
        "headline": "立法會將辯論施政報告",
        "headline_en": "LegCo to debate on Policy Address",
        "seq": "pr20251006-1   "
    }
}
]
```

'''

Please extract issue_date_en, headline, headline_en, and pdf_link_en for all news from 2024 to 2025 and save them in an Excel file.

I am a beginner in Python and have only installed Miniconda on Windows. You need to create a detailed readme.md tutorial (mainly to teach me how to use conda and python commands to start this project).

Suggestions for Vibe Coding for IT people



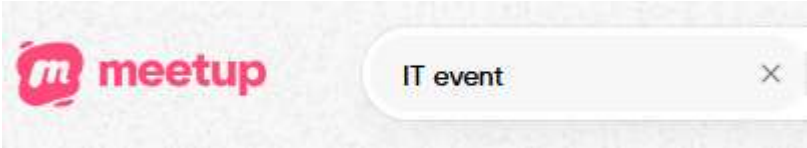
Reasons for one feature at a time :

1. AI can code better because it focuses more attention on this feature.
2. You can control the direction of iteration.
3. Human Unit Test and Code Review ensure code quality.

Part 3. My Open-source project by Vibe Coding

Stage 1 : Where is my open-source project idea from?

I like to join events after work, when I search on Meetup and click in the IT event that seems good.



Then the frustrated moment happened: I found the event is at Work Hours that I cannot join.

Then I started to think, could I develop a program to grab all the events from all platforms, and filters out those that occur during working hours or cover topics I'm not interested in, saving me time and effort.

But I didn't know Python frameworks (Python is the best language for Web Crawling) !

While, let me try AI – it helped me write 99% of the code!

Stage 2. Vibe Coding gave birth to my open-source project

Project GitHub

HK Event Search

python_hk_event_search allows you to search for Hong Kong events from various event platforms with a single command. It filters out events that occur during working hours and the topics you are not interested in, saving you time and effort.

Features

- Fetch events from Eventbrite, Luma, Meetup, GDG HK, HKU, CityU, PolyU, and HKBU.
- Filter for events during non-working hours, in person, free, not sold out, and the categories you are interested in.

Or search python_hk_event_search on GitHub

Three Common Web Crawling Methods used in this project

1.API

1. Access structured data (JSON) directly from a service's API.
2. Reliable and efficient.

2.HTML Parsing

1. Extract data from raw HTML.
2. Require handling page structure changes.

3.Browser

1. Initialize a browser and wait for dynamic content rendered by JavaScript to load.

EXPLORER

PYTHON_HK_EVENT_SEAR...

- __pycache__
- .vscode
  - settings.json
- .gitignore
- cityu_mba.py
- cityu.py
- common_utils.py
- environment.yml
- eventbrite.py
- gdg_hk.py
- hkbu.py
- hku.py
- luma.py
- main.py
- meetup.py
- polyu.py
- README.md

Welcome | README.md | luma.py

luma.py > fetch_events

```python
import time

def fetch_events():
    # https://luma.com/hongkong
    url = "https://api.lu.ma/discover/get-paginated-events?discover_place_api_id=discplace-z9B5Guglh2WINA1&pagination_limit=50"

    try:
        response = requests.get(url)
        if response.status_code != 200:
            print(f"Request error: {response.status_code}")
            return

        data = response.json()
        events = data.get("entries", [])

        for event in events:
            ticket_info = event.get("ticket_info", {})

            # Filter for events that are free, not sold out
            if ticket_info.get("is_free", False) and not ticket_info.get("is_sold_out", True):
                # Extract event details
                event_data = event.get("event", {})
                title = event_data.get("name", "")
                start_time_str = event_data.get("start_at", "")
                end_time_str = event_data.get("end_at", "")
```

1. API

```python
# Check if event is after work hours
if CommonUtils.is_period_after_work(event_start, event_end):
    print(f"Title: {title}")
    print(f"Start Date: {event_start.strftime('%Y-%m-%d %H:%M:%S')}, {event_start.strftime('%A')}")
    print(f"End Date  : {event_end.strftime('%Y-%m-%d %H:%M:%S')}, {event_end.strftime('%A')}")
    print(f"Venue: {venue}")
    print(f"URL: {event_url}")
    print("-" * 40)
```

2. HTML Parsing

EXPLORER

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- hkbu.py
- hku.py
- luma.py
- main.py M
- meetup.py
- polyu.py
- README.md

hku.py

hku.py > fetch_events

```python
# URL to scrape
url = "https://www.hku.hk/event/upcoming.html"

def fetch_events(title_filter_words):
    session = HTMLSession()
    response = session.get(url)

    # Parse the HTML content
    soup = BeautifulSoup(response.content, 'html.parser')
    events_table = soup.find('table', summary='a list of Upcoming Events')
    if not events_table:
        print("No events table found.")
        return

    events = events_table.find_all('tr')[1:]  # Skip header row

    for event in events:
        tds = event.find_all('td')
        if len(tds) < 4:
            # Not enough columns
            continue

        # Extract date and time
        date_time = tds[0].get_text(separator='\n', strip=True).split('\n')
        # Example: '25 Dec 2024 - 19 Jan 2025'
        date = date_time[0] if len(date_time) > 0 else ""
        # Example: '11:00-20:00'
        time = date_time[1] if len(date_time) > 1 else "00:00-00:00"

        # Extract start and end time
        if '-' in time:
            start_time, end_time = time.split('-', 1)
```

3. Browser

EXPLORER

PYTHON_HK_EVENT_SEAR...

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- hku.py
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- polyu.py
- README.md

OUTLINE

meetup.py M

meetup.py > ...

```python
from selenium import webdriver
from selenium.webdriver.chrome.options import Options
from selenium.webdriver.common.by import By
from selenium.webdriver.support.ui import WebDriverWait
from selenium.webdriver.support import expected_conditions as EC

def fetch_events(title_filter_words=[]):
    # categoryId=546 is for "Tech" events
    url = "https://www.meetup.com/find/?source=EVENTS&eventType=inPerson&sortField=DATETIME&location=hk--Kowloon&categoryId=546"

    driver = None
    try:
        # Set Chrome options
        chrome_options = Options()
        chrome_options.add_argument("--headless")  # Headless mode, browser not shown
        chrome_options.add_argument("--disable-gpu")
        chrome_options.add_argument("--no-sandbox")
        chrome_options.add_argument("--disable-dev-shm-usage")
        chrome_options.add_argument("--disable-blink-features=AutomationControlled")
        chrome_options.add_argument("--user-agent=Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko
        chrome_options.add_experimental_option("excludeSwitches", ["enable-automation"])
        chrome_options.add_experimental_option('useAutomationExtension', False)

        # Initialize Chrome WebDriver
        driver = webdriver.Chrome(options=chrome_options)
        driver.execute_cdp_cmd('Network.setUserAgentOverride', {"userAgent": 'Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebK

        # Visit the URL
        print("Loading the page with Selenium...")
        driver.get(url)

        # Wait for the page to load - increase wait time to ensure dynamic content is loaded
        print("Waiting for dynamic content to load...")
        # Wait for body to be present first
        wait = WebDriverWait(driver, 20)
        wait.until(EC.presence_of_element_located((By.TAG_NAME, "body")))
```

Ways to Filter the Topics

EXPLORER

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- polyu.py
- README.md

Welcome | README.md | luma.py | eventbrite.py M

eventbrite.py > ...

```python
import requests
import json
from datetime import datetime
import time
from common_utils import CommonUtils

url = "https://www.eventbrite.hk/api/v3/destination/search/"

#  Delete the categories you are not interested in
category_dict = {
    "Science & Tech": "EventbriteCategory/102",
    "Business": "EventbriteCategory/101",
    "Health": "EventbriteCategory/107",
    "Community": "EventbriteCategory/113",
    "Government": "EventbriteCategory/112",
    "Spirituality": "EventbriteCategory/114",

    "Charity & Causes": "EventbriteCategory/111",
    "Family & Education": "EventbriteCategory/115",
    "Fashion": "EventbriteCategory/106",
    "Film & Media": "EventbriteCategory/104",
    "Hobbies": "EventbriteCategory/119",
    "Home & Lifestyle": "EventbriteCategory/117",
    "Performing & Visual Arts": "EventbriteCategory/105",
    "School Activities": "EventbriteCategory/120",
    "Holidays": "EventbriteCategory/116",
    "Other": "EventbriteCategory/199",
}
```

main.py M

main.py > ...

```python
import cityu
import polyu
import eventbrite
import hku
import hkbu
import gdg_hk
import cityu_mba
import luma
import meetup  # Add this import for the new module

# Filter out topics by keywords in their titles
title_filter_config = {
    "hku": ["導賞","音樂節","中學","Endoscopy","Tubu
            ,"Infectious","Orthopaedic","Musculoske
    "hkbu": ["PhD programme","比賽","特展","崇拜","C
    "meetup": ["HKMechworks", "Coffee & Connections
    "eventbrite": ["新體驗","Public Social HackJam",
                   "Nook Bar","Networking Night","A
                   "Yogathon","Animal Flow","Ascens
}
```

Time Cost with Vibe Coding

- The first workable PoC (Proof of Concept) demo was completed in 10 mins
- The total project development and test time is no more than 3 hours

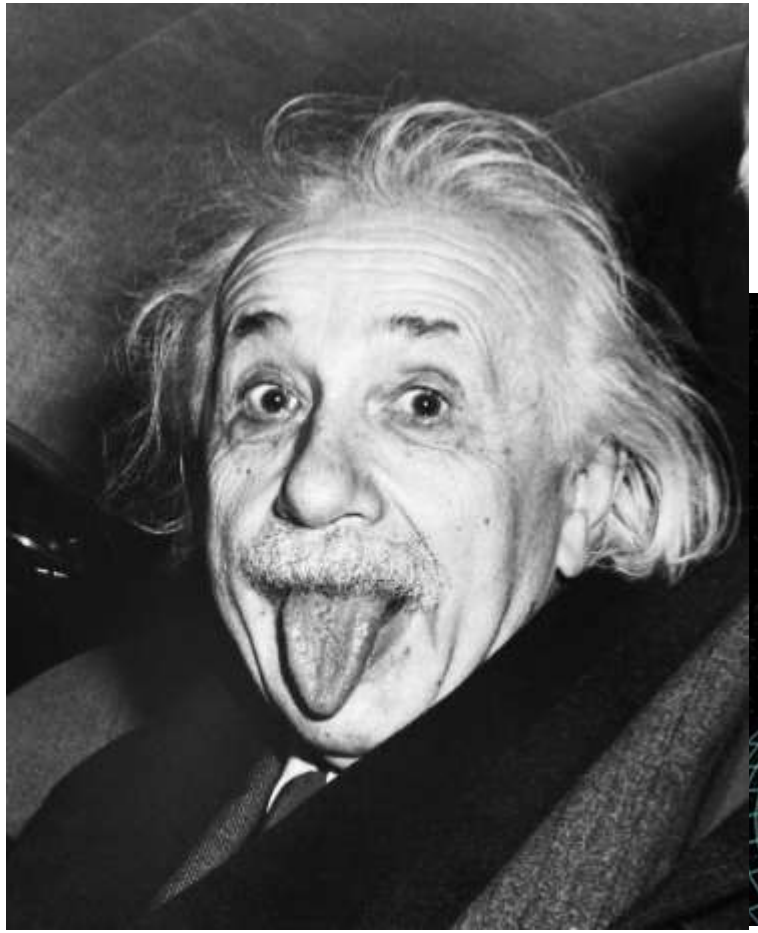
Time Cost without Vibe Coding

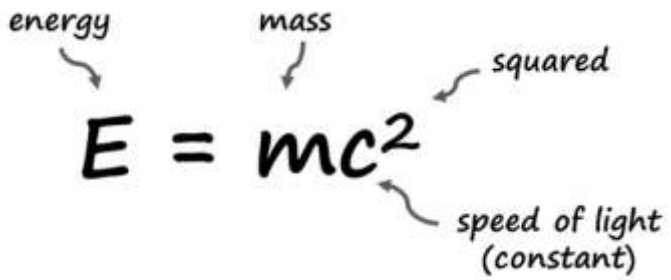
I don't have time to delve into Python web crawling technology and parse HTML patterns to turn my idea into an Open-source project.

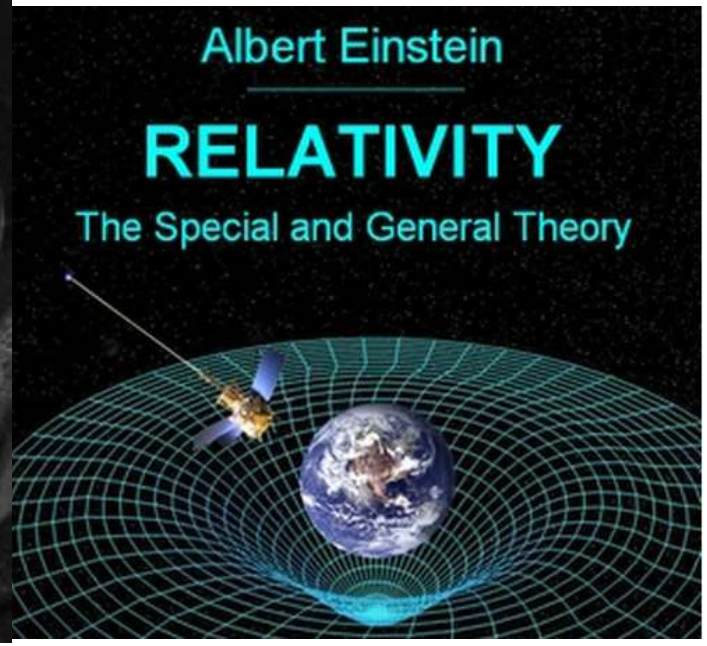
Question: If AI can code, then what about us Humans

AI is great for improving Productivity. But the current AI cannot create New things.

The new Ideas and Innovations rely on Humans.

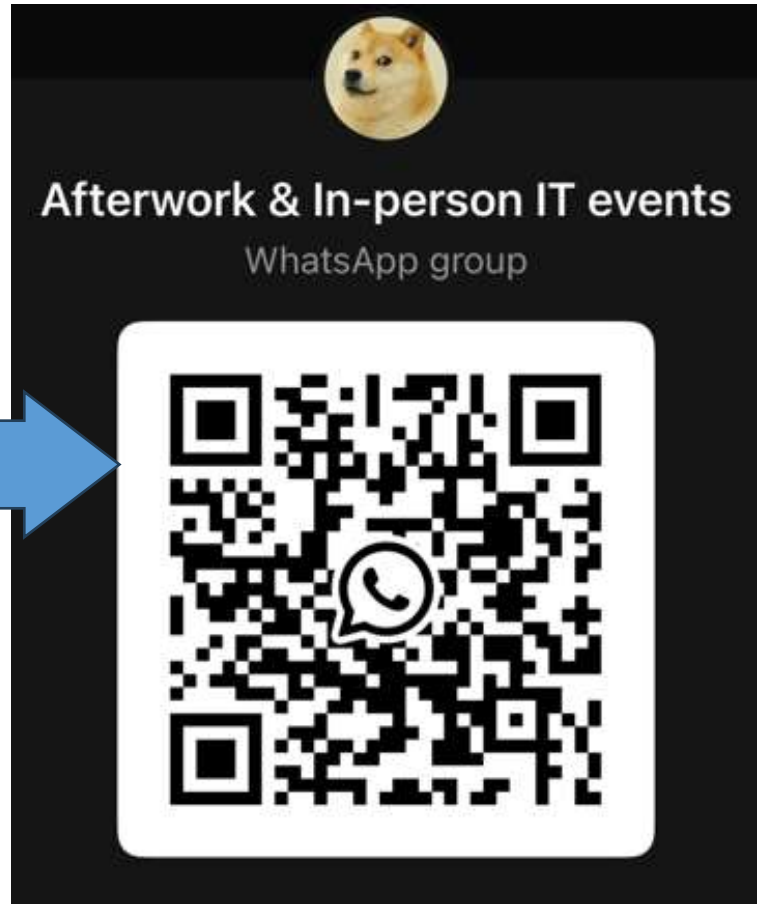
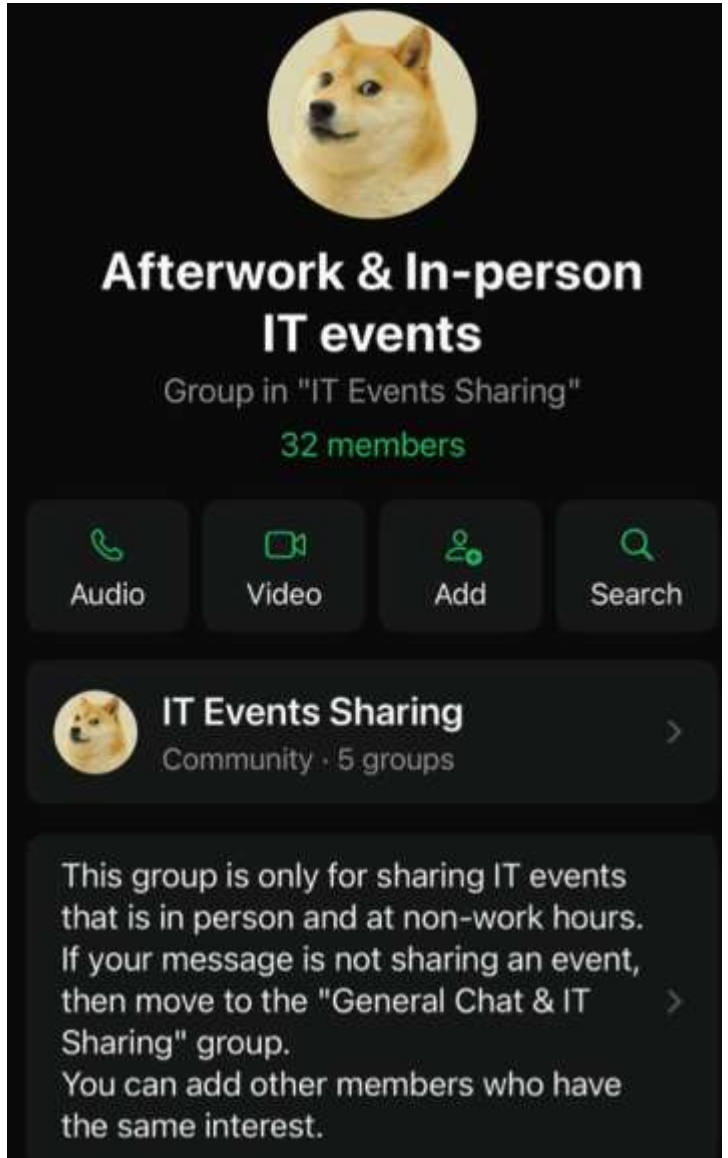
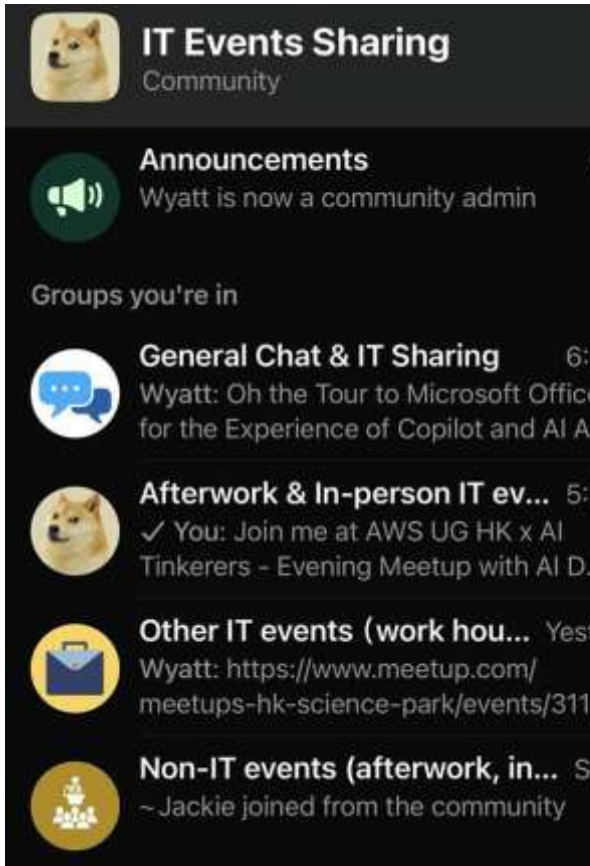






Stage 3. Ideas are from human, that's also why we have communities where more ideas are generated

After meeting more people who are also interested in IT events, I started thinking—why don't we have a community to share all events with everyone?

We have shared and will continue to share all the **Open Source Hong Kong** and **PyCon** events.

Welcome to join.